

Questions for the Audience

Assume a Two-layer Network with K Hidden and one Output Neurons

- How many learnable parameters does such a network have
 - $K \cdot (D + 1)$: For the first layer weight matrix $\mathbf{W}^{(1)}$
 - $K + 1$: For the first second layer weight $\vec{w}^{(2)}$
- How many cycles of a cosine function can we learn
 - $\frac{K}{2}$: with sigmoid (logistic/tanh) activation, one for each slope
- How many **global** minima does the loss function have?
 - $K!$: any permutation of neurons results in the same loss
 - 1: with a linear activation function

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How can you Speed up the Convergence of your Model?

- Select a reasonable (adaptive) learning rate
- Use (batched) stochastic gradient descent
- Provide more parallelized implementation
- Make use of modern hardware (MKL, GPU, TPU, CUDA)
- Reduce number of hidden units (can cause underfitting!)
- Momentum learning
- Apply other regularization techniques (see end of lecture)